

The misskappa package (draft)

`misskappa` is an R package for estimation and inference for weighted multi-rater agreement coefficients, including Conger's κ_C , Fleiss' κ_F , and the Brennan–Prediger coefficient. It supports raw rating matrices (subjects $\times$ raters) and count-form data, and allows missing ratings (NA).

Main user-facing functions.

- `kappa_raw(x, method, weight)`: categorical ratings in raw format (allows NA).
- `kappa_counts(x, method, weight, r)`: Fleiss-form counts (subjects $\times$ categories).
- `kappa_continuous(x, method, weight)`: continuous ratings (allows NA).
- `kappa(x, type = "auto", ...)`: convenience dispatcher.

Included example datasets.

- `dat.gwet2014`: raw $n \times R$ ratings with missing values (dim 20×5 ; NAs=22).
- `dat.klein2018`: raw $n \times R$ ratings with missing values (dim 10×5 ; NAs=3).
- `dat.zapf2016`: raw $n \times R$ ratings (complete) (dim 50×4 ; NAs=0).
- `dat.fleiss1971`: Fleiss-form counts (dim 30×5).

Example: missing ratings (`dat.gwet2014`)

Below is a minimal example (nominal/identity weights) using the three incomplete-data estimators used in the paper: available-case (AC), IPW, and Gwet.

```
library(misskappa)
data(dat.gwet2014)
x <- as.matrix(dat.gwet2014)

kappa_raw(x, method = "available", weight = "identity")
kappa_raw(x, method = "ipw", weight = "identity")
kappa_raw(x, method = "gwet", weight = "identity")
```

Table 1: Agreement estimates and asymptotic standard errors for `dat.gwet2014` (identity weights).

	$\hat{\kappa}_C$	SE	$\hat{\kappa}_F$	SE	$\hat{\kappa}_{BP}$	SE
AC	0.4845	0.0892	0.4780	0.0924	0.5014	0.0953
IPW	0.4233	0.1119	0.4183	0.1131	0.4421	0.1212
Gwet	0.4846	0.0899	0.4801	0.0919	0.5014	0.0953

Note. All `kappa_*` functions return a `misskappa_estimate` object with point estimates (`coef()`) and an asymptotic variance–covariance matrix (`vcov()`).